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B.Tech. DEGREE EXAMINATION, MAY 2022

Third Semester

18MES201T – ENGINEERING MECHANICS

(For the candidates admitted from the academic year 2018-2019 to 2019-2020)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- (ii) **Part - B** should be answered in answer booklet.

Time: 2½ Hours

Max. Marks: 75

PART – A (25 × 1 = 25 Marks)

Answer ALL Questions

- | | Marks | BL | CO | PO |
|--|-------|----|----|----|
| 1. Any object whose dimension is not used in analysis, then it is assumed as
(A) Particle (B) Rigid body
(C) Plastic body (D) Deformable body | 1 | 1 | 1 | 1 |
| 2. Couple is a _____ vector.
(A) Sliding (B) Fixed
(C) Null (D) Free | 1 | 2 | 1 | 1 |
| 3. Freely sliding collar is an example of _____ support.
(A) Fixed (B) Hinged
(C) Roller (D) Welded | 1 | 1 | 1 | 2 |
| 4. The forces which meet at one joint but their lines of action donot lie in a plane are called
(A) Coplanar non-concurrent forces (B) Non coplanar concurrent forces
(C) Non coplanar non concurrent forces (D) Intersecting forces | 1 | 2 | 1 | 2 |
| 5. According to principle of transmissibility of forces, the effect upon a body is
(A) Maximum when it acts at the CG (B) Different at different points in its line of action
(C) Same at every point in its line of action (D) Minimum when it acts at the CG of the body | 1 | 1 | 1 | 1 |
| 6. The force of friction acts in a direction _____ to the direction of motion of object.
(A) Same (B) Opposite
(C) Perpendicular (D) Downwards | 1 | 2 | 2 | 1 |
| 7. The coefficient of friction depends on
(A) Area of contact (B) Strength of surface
(C) Nature of surface (D) Shape of contact area | 1 | 1 | 2 | 2 |

8. The ratio of the limiting force of friction (F) to the normal reaction (R) is known as 1 2 2 1
 (A) Coefficient of friction (B) Force of friction
 (C) Angle of friction (D) Limiting friction
9. Friction force encountered after commencement of motion is called 1 1 2 2
 (A) Dynamic friction (B) Static friction
 (C) Dry friction (D) Angle of friction
10. When a ladder resting on the smooth ground and leaning against the vertical wall, the force of friction will be 1 2 2 1
 (A) Downwards at its upper end (B) Upwards at its upper end
 (C) Zero at its upper end (D) Perpendicular to the wall at its upper end
11. The point through which the entire weight of the body acts, irrespective of its position is 1 2 3 2
 (A) Mass moment of inertia (B) Moment of inertia
 (C) Centre of gravity (D) Centre of mass
12. The centre of gravity of a semi circle lies at a distance of _____ from its base measured along the vertical radius. 1 1 3 1
 (A) $4R/3\pi$ (B) $3R/4\pi$
 (C) $3R/5\pi$ (D) $5R/3\pi$
13. The centre of gravity of right circular solid cone of height 'h' lies at _____ distance from the base. 1 2 3 2
 (A) $h/2$ (B) $h/6$
 (C) $h/4$ (D) $h/3$
14. Centroid of a straight line having a length 'L' lies at 1 1 3 1
 (A) L (B) $L/2$
 (C) $L/3$ (D) $L/4$
15. The moment of inertia of a hollow cylinder section with outer diameter, D and inner diameter d is 1 2 3 2
 (A) $\frac{\pi}{64}(D^4 - d^4)$ (B) $\frac{\pi}{32}(D^4 - d^4)$
 (C) $\frac{\pi}{64}(D^2 - d^2)$ (D) $\frac{\pi}{32}(D^2 - d^2)$
16. _____ is the total distance covered in total duration. 1 1 4 1
 (A) Uniform velocity (B) Average velocity
 (C) Average speed (D) Variable speed
17. _____ is the study of bodies subjected to unbalanced forces. 1 2 4 2
 (A) Kinetics (B) Kinematics
 (C) Dynamics (D) Statics
18. The motion along a straight path is called 1 1 4 1
 (A) Rectilinear motion (B) Curvilinear motion
 (C) Average velocity motion (D) Instantaneous velocity

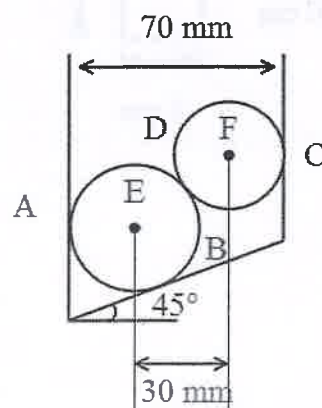
19. The motion of a particle along a straight line is given by $x = t^3 + 3.5t^2 + 7$, then its acceleration is equal to
 (A) $6t + 7$ (B) $3t^2 + 7t$
 (C) $6t - 7$ (D) $3t^2 + 7$
20. The coefficient of restitution for plastic and elastic impact is
 (A) 1 and 0 (B) $1 < e < 2$
 (C) $0 < e < 1$ (D) 0 and 1
21. When the distance between any two points do not change due to external forces is known as
 (A) Particle (B) Rigid body
 (C) Elastic body (D) Projectile
22. Rolling wheel is an example of
 (A) Linear translation (B) Rotation about fixed point
 (C) General plane motion (D) Rigid body rotation
23. If $w_0 = 209.4 \text{ rad/sec}$, $w = 0$ and time taken is 100 sec. Calculate $\alpha = ?$
 (A) 2.094 rad/s^2 (B) -2.094 rad/s^2
 (C) 309.4 rad/s^2 (D) -309.4 rad/s^2
24. In the engine, crank rotates at 400 rpm with constant angular velocity. Determine the angular velocity of the crank.
 (A) 418 rad/s (B) 518 rad/s
 (C) 320 rad/s (D) 618 rad/s
25. The rate of change of angular momentum in terms of angular acceleration is
 (A) $I\alpha$ (B) I/α
 (C) α/I (D) IW

PART – B ($5 \times 10 = 50$ Marks)

Answer ALL Questions

Marks BL CO PO

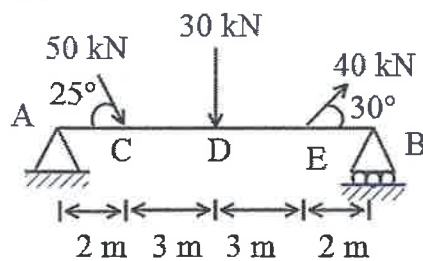
26. a. Two cylinders of diameters 60 mm and 35 mm weighing 170 N and 50 N respectively are placed as shown in figure. Assuming all surfaces are smooth, find the reactions at A, B and C.



(OR)

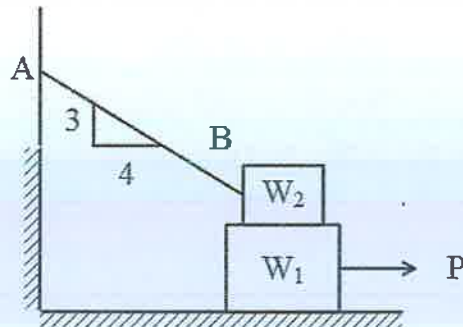
b. Find the reactions at support A and B

10 4 1 2



27. a. A block of weight $W_1 = 1500 \text{ N}$ rests on a horizontal surface and supports another block of weight $W_2 = 600 \text{ N}$ on top of it. A string AB is attached to block of weight W_2 . Find the force 'P' required to cause slipping to impend. Take coefficient of friction as 0.3 for all surface.

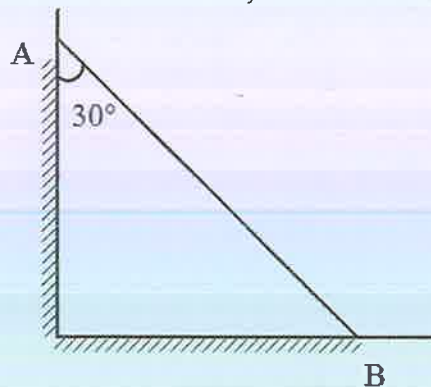
10 4 2 1



(OR)

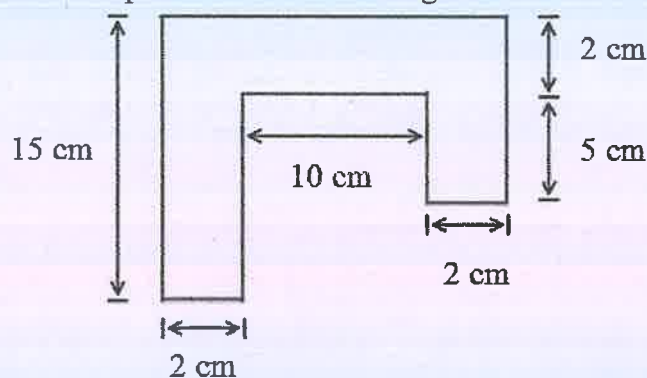
- b. A ladder of weight 400 N and 6 m long is placed against a wall as shown in figure. The coefficient of friction is 0.4 for all contact surfaces. Find how high a man of weight 1500 N can ascend, before the ladder begins to slip.

10 3 2 2



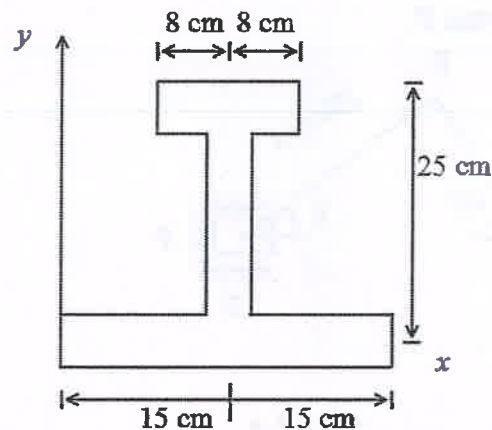
28. a. Locate the centroid of plane area shown in figure.

10 4 3 1



(OR)

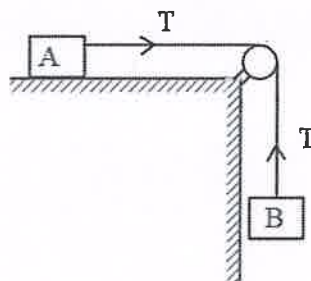
- b. Find I_{xx} and I_{yy} through centroid axis with a uniform thickness of 3 cm through out. 10 3 3 2



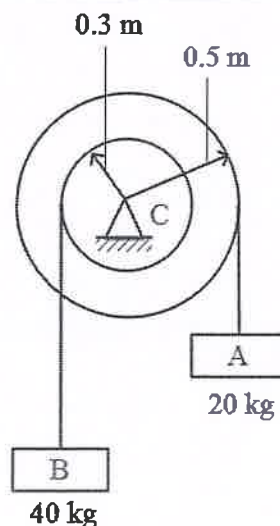
29. a. A shot is fired with a bullet with an initial velocity of 50 m/s from a point 20 m in front of a vertical wall 10 m high. Find the angle of projection with horizontal to enable the bullet to just clear the wall. 10 4 4 1

(OR)

- b. A body of mass 30 kg resting on a horizontal table as shown in figure. It is connected by a string over a smooth pulley to another mass 5 kg. Initially, the friction is just sufficient to prevent the motion. If an additional 2 kg is added to 5 kg mass, find the acceleration of the masses. 10 3 4 2



30. a. A 150 kg pulley having radius of gyration of 0.4 m is connected to two cylinders as shown in figure. Assume no friction and determine the angular acceleration of the pulley and acceleration of each cylinder. 10 3 5 1



(OR)

- b. In the engine system, the crank AB has a constant angular velocity of 5000 rpm. For the crank position indicated, determine
- The angular velocity of connecting rod BD
 - The velocity of piston P

